`GeoModels` Tutorial: simulation, estimation and prediction of spatial data with asymmetric and heavy tails using sinh-arcsinh (SAS) random fields

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Introduction

In this tutorial we show how to analyze spatial data with flexible tails using sinh-arcsinh (SAS) random fields (Blasi et al., 2022) with the R package `GeoModels` (Bevilacqua et al., 2026). This process incorporates flexible marginal distributions involving two additional parameters that model heavier or lighter tails than those induced by Gaussian processes and/or possible asymmetries. One advantage of the SAS compared to other recent proposals (e.g., Xu and Genton (2017)) is that the transformation involved is explicitly invertible, which allows likelihood-based methods of estimation to be applied directly.

We first load the R libraries needed in this tutorial and set the name of the model in the `GeoModels` package. The code assumes that the package has already been installed.

```
rm(list=ls());
require(GeoModels);
require(fields);
model="SinhAsinh"; # model name in the GeoModels package
set.seed(91);
```

Simulation of sinh-arcsinh (SAS) random fields

The definition of a SAS random field starts by considering a parent Gaussian random field $G = \{G(\boldsymbol{s}), \boldsymbol{s} \in D\}$, where $\boldsymbol{s}$ represents a location in the domain D. In this tutorial we consider the spatial case, $D \subseteq \mathbb{R}^2$. However, `GeoModels` also allows spatio-temporal data or data defined on a sphere of arbitrary radius. The Gaussian field G is assumed weakly stationary with zero mean, unit variance and correlation function $\rho(\boldsymbol{h}) = \text{cor}\{G(\boldsymbol{s}+\boldsymbol{h}), G(\boldsymbol{s})\}$.

Let us consider the continuous, strictly monotonic function $S_{a,b} : \mathbb{R} \to \mathbb{R}$ defined as

$$S_{a,b}(y) = \sinh\{b \,\text{asinh}(y) - a\}\,, \qquad b > 0,\ a \in \mathbb{R}. \tag{1}$$

Specifically, we define $Y^*_{\eta,\kappa} = \{Y^*_{\eta,\kappa}(\boldsymbol{s}), \boldsymbol{s} \in D\}$ by $Y^*_{\eta,\kappa}(\boldsymbol{s}) = S^{-1}_{\eta,\kappa}\{G(\boldsymbol{s})\}$, where $S^{-1}_{\eta,\kappa}(z) = \sinh\{[\text{asinh}(z) + \eta]/\kappa\}$. One particularly appealing property of the SAS transformation is that its parameters are interpretable. The parameter $\eta \in \mathbb{R}$ controls skewness: $\eta > 0$ and $\eta < 0$ yield right and left skewness, respectively. The parameter $\kappa > 0$ controls tail weights:

$\kappa < 1$ and $\kappa > 1$ yield heavier and lighter tails than the Gaussian distribution, respectively. The case $\eta = 0$ and $\kappa = 1$ gives the Gaussian random field.

The standardized SAS RF has marginal density (Jones and Pewsey, 2009)

$$f_{Y^*_{\eta,\kappa}}(y) = \frac{\kappa}{\sqrt{2\pi}} \left\{ \frac{1 + S^2_{\eta,\kappa}(y)}{1 + y^2} \right\}^{1/2} \exp\left\{ -\frac{1}{2} S^2_{\eta,\kappa}(y) \right\}. \tag{2}$$

Its mean $m_{\eta,\kappa} = \mathrm{E}\{Y^*_{\eta,\kappa}(\boldsymbol{s})\}$ and variance $v_{\eta,\kappa} = \mathrm{Var}\{Y^*_{\eta,\kappa}(\boldsymbol{s})\}$ are

$$\begin{aligned} m_{\eta,\kappa} &= \frac{e^{1/4} \sinh(\eta/\kappa)}{\sqrt{8\pi}} \left\{ K_{(\kappa+1)/(2\kappa)}(1/4) + K_{(1-\kappa)/(2\kappa)}(1/4) \right\}, \\ v_{\eta,\kappa} &= \frac{e^{1/4} \cosh(2\eta/\kappa)}{\sqrt{32\pi}} \left\{ K_{(\kappa+2)/(2\kappa)}(1/4) + K_{(2-\kappa)/(2\kappa)}(1/4) \right\} - \frac{1}{2} - m^2_{\eta,\kappa}, \end{aligned}$$

where K_ζ is the modified Bessel function of the second kind of order ζ. Since the random field $Y^*_{\eta,\kappa}$ is standardized with respect to the location-scale transformation below, its mean does *not* involve the scale parameter σ.

The correlation function of the SAS random field admits the expansion

$$\rho_{Y^*_{\eta,\kappa}}(\boldsymbol{h}; \boldsymbol{\psi}) = \sum_{j=1}^{\infty} \frac{\xi_j(\eta,\kappa)^2}{j!} \rho(\boldsymbol{h}; \boldsymbol{\psi})^j, \tag{3}$$

where the Hermite coefficients ξ_j are normalized so that $\sum_{j \geq 1} \xi_j(\eta,\kappa)^2/j! = 1$. A closed form for these coefficients is provided by Blasi et al. (2022).

A location and scale transformation gives the nonstandard SAS random field

$$Y_{\eta,\kappa}(\boldsymbol{s}) = \mu(\boldsymbol{s}) + \sigma Y^*_{\eta,\kappa}(\boldsymbol{s}), \tag{4}$$

where $\mu(\boldsymbol{s}) = X(\boldsymbol{s})^\top \boldsymbol{\beta}$ is the spatially varying *location* parameter and $\sigma > 0$ is the scale parameter. The marginal mean is $\mathrm{E}\{Y_{\eta,\kappa}(\boldsymbol{s})\} = \mu(\boldsymbol{s}) + \sigma m_{\eta,\kappa}$, and its marginal variance is $\sigma^2 v_{\eta,\kappa}$. The first entry of the k-dimensional vector $X(\boldsymbol{s})$ is one for the intercept. In this example we assume $k = 2$.

To obtain a realization from $Y_{\eta,\kappa}$ we specify two regression coefficients, the skewness and tail parameters, the scale σ^2, and the parent correlation. We first set the spatial coordinates:

```
N=1500;
coords=cbind(runif(N),runif(N));
plot(coords,pch=20,xlab="",ylab="");
```

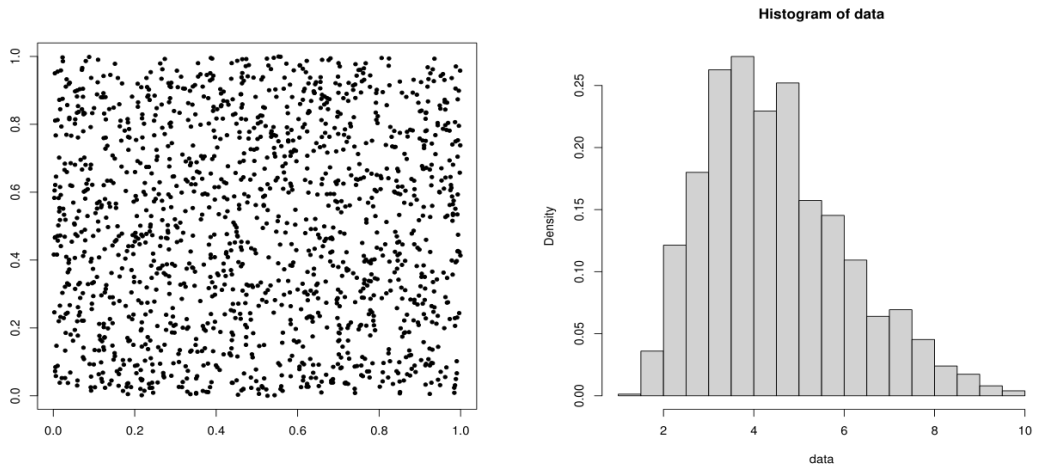


Figure 1: From left to right: spatial location sites and histogram of a realization of the SAS random field. Original figure (2025).

For the correlation function $\rho(\boldsymbol{h})$ of the parent Gaussian random field we assume an isotropic Matérn model (Matérn, 1986):

$$\rho_{\alpha,\gamma}(\boldsymbol{h}) = \frac{2^{1-\gamma}}{\Gamma(\gamma)} \left(\frac{\|\boldsymbol{h}\|}{\alpha} \right)^{\gamma} K_{\gamma} \left(\frac{\|\boldsymbol{h}\|}{\alpha} \right), \qquad \|\boldsymbol{h}\| \geq 0, \tag{5}$$

where $\gamma > 0$ is the smoothness parameter and $\alpha > 0$ the spatial scale parameter. We set the associated parameters:

```
corrmodel="Matern";
scale=0.15/3;
smooth=0.5;
nugget=0;
```

We set skew, tail and scale parameters of the SAS random field:

```
skew=3.5; # skewness parameter
tail=2; # tail weight (>1: lighter-than-Gaussian tails)
sill=1.5; # sigma^2 (not the marginal variance)
```

Finally, the location regression parameters and the regression matrix are

```
mean=1; mean1=-1;
a0=rep(1,N); a1=runif(N,-1,1);
X=cbind(a0,a1);
```

We simulate a realization using `GeoSim`. The parent Gaussian field is generated by Cholesky decomposition; approximate simulation methods, such as circulant embedding and turning bands, are also available through `GeoSimApprox` (Bevilacqua et al., 2026).

```
param=list(nugget=nugget,mean=mean,mean1=mean1,
          scale=scale,smooth=smooth,sill=sill,skew=skew,tail=tail);
data <- GeoSim(coordx=coords,corrmodel=corrmodel,
                        param=param,model=model,X=X)$data;
hist(data,prob=TRUE,breaks=20);
```

The simulated data exhibit asymmetry. For $\kappa = 2$ the marginal tails are lighter than Gaussian tails, although the SAS family also accommodates heavier tails for $\kappa < 1$.

Estimation of SAS random fields

Given $\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\} \subset D$ distinct sites, let $\boldsymbol{\Omega}(\boldsymbol{\psi}) = [\rho(\|\boldsymbol{s}_i - \boldsymbol{s}_j\|; \boldsymbol{\psi})]_{i,j=1}^n$ be the correlation matrix of the parent Gaussian field. Maximum likelihood estimation can be performed using `GeoFit` or `GeoFit2`. Define $u_i = (y_i - \mu_i)/\sigma$ and $z_i = S_{\eta,\kappa}(u_i)$, and set $\boldsymbol{z} = (z_1, \ldots, z_n)^\top$. The log-likelihood associated with $\boldsymbol{Y} = (Y_{\eta,\kappa}(\boldsymbol{s}_1), \ldots, Y_{\eta,\kappa}(\boldsymbol{s}_n))^\top$ is

$$\begin{aligned}\ell(\boldsymbol{\vartheta}; \boldsymbol{y}) = & -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2}\log\det\boldsymbol{\Omega}(\boldsymbol{\psi}) - \frac{1}{2}\boldsymbol{z}^\top\boldsymbol{\Omega}(\boldsymbol{\psi})^{-1}\boldsymbol{z} + n\log\kappa \\ & + \frac{1}{2}\sum_{i=1}^{n}\left[\log(1+z_i^2) - \log(1+u_i^2)\right],\end{aligned} \tag{6}$$

where $\boldsymbol{\vartheta} = (\boldsymbol{\beta}^\top, \boldsymbol{\psi}^\top, \sigma, \eta, \kappa)^\top$. The sign preceding the last sum is *positive*, as follows from the Jacobian $\mathrm{d}S_{\eta,\kappa}(u)/\mathrm{d}u = \kappa\{(1 + S_{\eta,\kappa}(u)^2)/(1+u^2)\}^{1/2}$. If $\eta = 0$ and $\kappa = 1$, (6) reduces to the Gaussian log-likelihood. The *maximum likelihood* estimator maximizes (6); it is not a pairwise likelihood estimator.

Maximum likelihood estimation is performed using `GeoFit2`. We use `nlminb`, which allows box-constrained optimization; other optimizers can be used. The constraints on strictly positive parameters exclude zero:

```
optimizer="nlminb";
fixed<-list(nugget=nugget,smooth=smooth);
start<-list(mean=mean,mean1=mean1,scale=scale,sill=sill,
           skew=skew,tail=tail);
```

```
I=Inf;
lower<-list(mean=-I,mean1=-I,scale=1e-6,sill=1e-6,
           skew=-I,tail=1e-6);
upper<-list(mean=I,mean1=I,scale=I,sill=I,skew=I,tail=I);
fit1 <- GeoFit2(data=data,coordx=coords,
   corrmodel=corrmodel,X=X,model=model,optimizer=optimizer,
   lower=lower,upper=upper,likelihood="Full",type="Standard",
   varest=TRUE,start=start,fixed=fixed);
fit1;
```

The original tutorial reported the following full maximum likelihood estimates and standard errors. These values are retained as historical output and have not been recomputed with `GeoModels` 2.3.0:

```
# full maximum likelihood
# mean mean1 scale sill skew tail
# estimates 2.04476 -1.00415 0.04624 1.71296 1.98039 1.53782
# std. error 0.380464 0.021201 0.004139 0.424051 0.478868 0.193636
# log-likelihood = -1857.18; AIC = 3726; BIC = 3758
```

An alternative, computationally faster method is weighted composite likelihood based on marginal pairs. Denote the log density of the bivariate random vector $\{Y_{\eta,\kappa}(\boldsymbol{s}_i), Y_{\eta,\kappa}(\boldsymbol{s}_j)\}^\top$ by $\ell_{ij}(\boldsymbol{\vartheta}; \boldsymbol{y})$. The weighted pairwise log-likelihood is

$$\mathrm{pl}(\boldsymbol{\vartheta}) = \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \ell_{ij}(\boldsymbol{\vartheta}; \boldsymbol{y}) w_{ij}, \tag{7}$$

where w_{ij} are nonnegative weights independent of $\boldsymbol{\vartheta}$. Following Caamaño-Carrillo et al. (2024), one specification based on nearest neighbors is

$$w_{ij}(k) = \begin{cases} 1, & \boldsymbol{s}_i \in N_k(\boldsymbol{s}_j), \\ 0, & \text{otherwise}, \end{cases} \tag{8}$$

where $N_k(\boldsymbol{s})$ is the set of k nearest neighbors. The pairwise likelihood estimator maximizes (7), rather than (6). The parameter vector may be written $(\boldsymbol{\beta}^\top, \boldsymbol{\psi}^\top, \sigma^2, \eta, \kappa)^\top$ when working with `sill` instead of σ.

Pairwise likelihood estimation can be performed using `GeoFit2`. Here we set `neighb=2`, corresponding to $k = 2$ in (8):

```
fit2 <- GeoFit2(data=data,coordx=coords,
   corrmodel=corrmodel,X=X,model=model,optimizer=optimizer,
   lower=lower,upper=upper,neighb=2,
   likelihood="Marginal",type="Pairwise",sensitivity=TRUE,
   start=start,fixed=fixed);
fit2;
```

The original pairwise composite likelihood estimates were:

```
# marginal pairwise likelihood
# mean mean1 scale sill skew tail
# estimates 2.18162 -0.98421 0.04628 1.75475 1.77451 1.47713
# maximum composite log-likelihood = -8884.95
```

Standard errors can also be obtained by parametric bootstrap using `GeoVarestbootstrap`. We preserve $K = 500$ as in the original example:

```
fit2=GeoVarestbootstrap(fit2,K=500,parallel=TRUE);
round(fit1$stderr,5); round(fit2$stderr,5);
```

The historical pairwise standard errors were 0.40349 (mean), 0.02291 (mean1), 0.00466 (scale), 0.60544 (sill), 0.60791 (skew) and 0.21759 (tail).

The *full ML confidence intervals* printed in the original PDF were much narrower than those implied by the reported standard errors. For consistency, we instead compute two-sided 95% Wald intervals from the estimated parameters and standard errors:

```
wald_ci <- function(fit,level=0.95) {
  estimates <- unlist(fit$param)
  errors <- unlist(fit$stderr)
  common <- intersect(names(estimates),names(errors))
  if (!length(common)) stop("Missing matching SEs")
  z <- qnorm((1+level)/2)
  cbind(lower=estimates[common]-z*errors[common],
        upper=estimates[common]+z*errors[common])
}
round(wald_ci(fit1),5); round(wald_ci(fit2),5);
```

The following are Wald 95% intervals recalculated *from the rounded historical ML estimates*

and standard errors, not from a new execution:

	mean	mean1	scale	sill	skew	tail
Lower	1.29907	-1.04570	0.03813	0.88184	1.04183	1.15830
Upper	2.79046	-0.96260	0.05435	2.54409	2.91895	1.91734

In the historical example the reported ML standard errors were slightly smaller than the pairwise likelihood standard errors for these parameters.

Checking model assumptions

Hereafter we use the full ML fit for diagnostics and prediction. With the location regression coefficients and the scale parameter, the estimated residuals

$$\widehat{Y^*_{\eta,\kappa}}(\boldsymbol{s}_i) = \frac{y(\boldsymbol{s}_i) - X(\boldsymbol{s}_i)^\top \widehat{\boldsymbol{\beta}}}{\widehat{\sigma}}, \qquad i = 1, \ldots, N,$$

can be viewed as a realization of the standardized SAS process. Such residuals are *not necessarily mean-zero* when $\eta \neq 0$. They can be computed using `GeoResiduals`:

```
res=GeoResiduals(fit1);
GeoQQ(res);
vario <- GeoVariogram(data=res$data,
                              coordx=coords,maxdist=0.4);
GeoModels::GeoCovariogram(res,show.vario=TRUE,vario=vario,pch=20);
```

The marginal distribution is examined with a QQ plot and dependence with the empirical and fitted semivariograms (Figure 2). The theoretical correlation is obtained from (3).

Prediction of SAS random fields

For a location $\boldsymbol{s}_0$ with covariates $X(\boldsymbol{s}_0)$, the optimal *linear* prediction of the SAS random field is

$$\widehat{Y}_{\eta,\kappa}(\boldsymbol{s}_0) = m(\boldsymbol{s}_0) + \sum_{i=1}^{N} \lambda_i \{y(\boldsymbol{s}_i) - m(\boldsymbol{s}_i)\}, \qquad m(\boldsymbol{s}) = X(\boldsymbol{s})^\top \boldsymbol{\beta} + \sigma m_{\eta,\kappa}, \tag{9}$$

where $m_{\eta,\kappa} = \mathrm{E}(Y^*_{\eta,\kappa})$ is defined above, and $\boldsymbol{\lambda} = R^{-1}_{\eta,\kappa} \boldsymbol{c}_{\eta,\kappa}$ with

- $\boldsymbol{c}_{\eta,\kappa} = (\mathrm{cor}\{Y_{\eta,\kappa}(\boldsymbol{s}_0), Y_{\eta,\kappa}(\boldsymbol{s}_1)\}, \ldots, \mathrm{cor}\{Y_{\eta,\kappa}(\boldsymbol{s}_0), Y_{\eta,\kappa}(\boldsymbol{s}_N)\})^\top$;

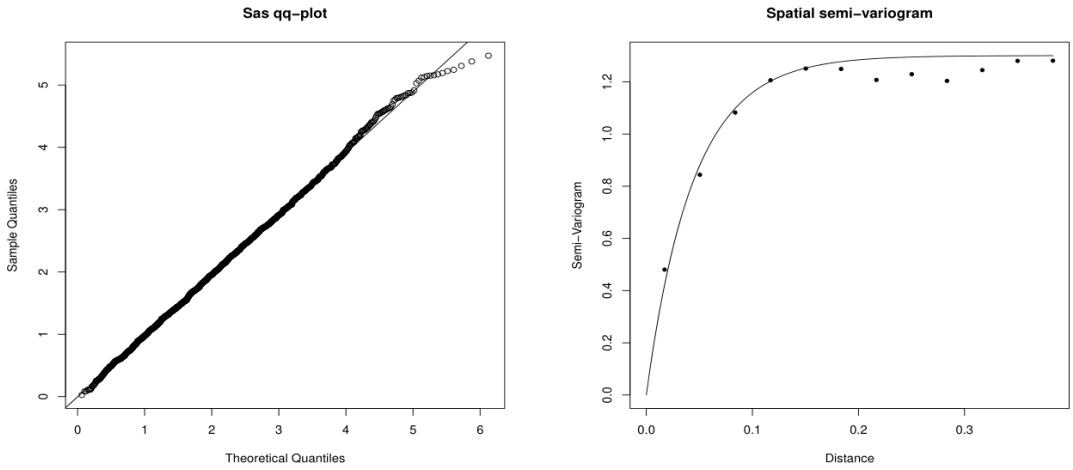


Figure 2: From left to right: SAS QQ plot of the standardized residuals and empirical versus fitted semivariogram. Original 2025 figure; the reported fitted shape parameters are approximately $\eta = 1.98$ and $\kappa = 1.54$.

- $R_{\eta,\kappa} = [\text{cor}\{Y_{\eta,\kappa}(\boldsymbol{s}_i), Y_{\eta,\kappa}(\boldsymbol{s}_j)\}]_{i,j=1}^{N}$.

Because $m_{\eta,\kappa}$ is generally nonzero when $\eta \neq 0$, the location parameter $X(\boldsymbol{s})^\top \boldsymbol{\beta}$ cannot be substituted for $m(\boldsymbol{s})$ in (9). The mean square error (MSE), conditional on the fitted model parameters, is

$$\text{MSE}\{\widehat{Y}_{\eta,\kappa}(\boldsymbol{s}_0)\} = \sigma^2 v_{\eta,\kappa} \left(1 - \boldsymbol{c}_{\eta,\kappa}^\top R_{\eta,\kappa}^{-1} \boldsymbol{c}_{\eta,\kappa}\right). \tag{10}$$

If parameters are unknown, we use the fitted values in (9)–(10). The correlation quantities are computed using (3) and fitted Matérn, skewness and tail parameters (not degrees of freedom).

Kriging and its associated MSE can be obtained using `GeoKrig`. We first set the regular prediction grid and the associated covariates:

```
xx=seq(0,1,0.012);
loc_to_pred=as.matrix(expand.grid(xx,xx));
Nloc=nrow(loc_to_pred);
Xloc=cbind(rep(1,Nloc),runif(Nloc));
pr=GeoKrig(fit1,loc=loc_to_pred,
                      Xloc=Xloc,mse=TRUE);
```

As in the original simulation, prediction covariates are generated independently at grid sites; thus the resulting map represents both spatial dependence and covariate variation. The following displays the simulated field, kriging map and corresponding MSE:

```
par(mfrow=c(1,3));
quilt.plot(coords[,1],coords[,2],data,main="Data");
map=matrix(pr$pred,ncol=length(xx));
image.plot(xx,xx,map,xlab="",ylab="",main="SimpleKriging");
map_mse=matrix(pr$mse,ncol=length(xx));
image.plot(xx,xx,map_mse,xlab="",ylab="",main="MSE");
```

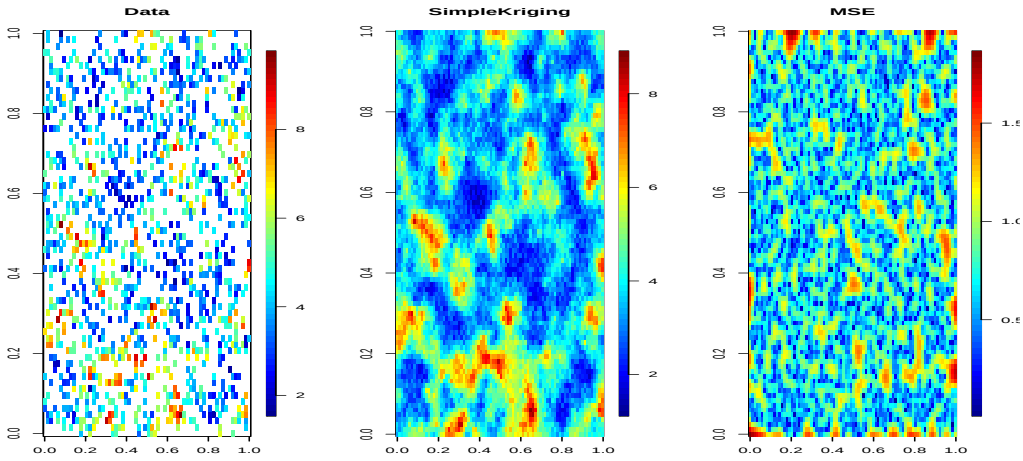


Figure 3: From left to right: observed spatial data, associated kriging map and mean squared error map.

References

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